

Seven-Colour LED Lighting

RIJU THAZHATHUVEETIL

Presented here is a simple circuit that uses six red, green and blue (RGB) LEDs to generate a running-light effect in seven colours – blue, green, red, cyan, yellow, magenta and white. The RGB LEDs have four pins, one for each colour and a com-

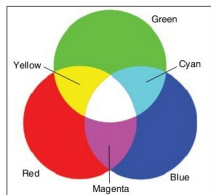


Fig. 1: Colours produced by three basic colours

mon cathode. Different colours can be produced by mixing the primary colours (i.e., red, green and blue).

Circuit and working

The circuit diagram for the seven-colour LED lighting is shown in Fig. 2. The circuit is built around timer NE555 (IC1), decade counter CD4017B (IC2), four-bit binary counter 7493 (IC3), driver IC ULN2003 (IC4), six

RGB LEDs (RGB1 through RGB6), three BC547 transistors (T1 through T3) along with some basic components.

The timer IC NE555 (IC1) is wired in astable multivibrator mode. It provides low-frequency clock pulses, the frequency of which is decided by timing components R1, VR1 and C2. The frequency of the timer can be varied using the preset VR1.

TABLE I
The BCD Output of IC3

Count	QB	QC	QD	RGB1-RGB6
1	H	L	L	Blue colour selected
2	L	H	L	Green colour selected
3	L	L	H	Red colour selected
4	H	H	L	Cyan colour selected
5	L	H	H	Yellow colour selected
6	H	L	H	Magenta colour selected
7	H	H	H	White colour selected

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Clock pulse for IC2 (adjusted using VR1)
TP3	Clock pulse for the binary counter (frequency depends on the frequency of timer IC1)

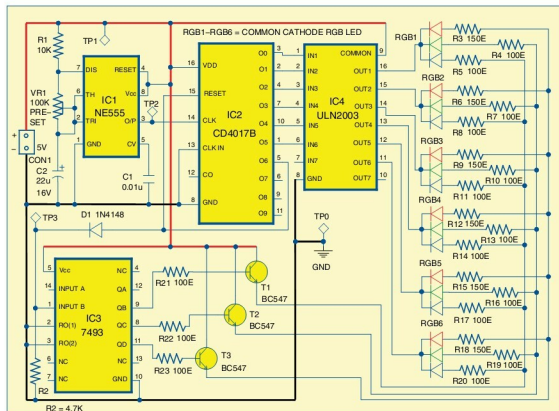


Fig. 2: Circuit of seven-colour LED lighting

The output of IC1 at pin 3 is fed to decade counter CD4017B (IC2). IC2 has ten outputs, which go high in sequence when a source of pulses is connected to its CLK input (pin 14). Q0 through Q5 of IC2 are connected to IN1 through IN6 of IC4. Pin Q6 of IC2 is connected back to its reset pin 15 and input pin 1 of IC3 via diode D1. When Q6 pin of IC2 goes high, it resets IC2 and, at the same time, provides the